



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

UNIVERSITY INSTITUTE OF COMPUTING (UIC)

ProjectReport ON Guessing Game C++

Program Name: BCA Subject

**Name/Code: Object Oriented
Programming(24CAH-201)**

Submitted by: Name:

Harshdeep Kaur

UID: 24BCA10370

Section: 24BCA 2

Group: B

Submitted to:

Name: Ms. RashmiSaluja

Designation: AssistantProfessor

Table of Contents

Acknowledgement	3
Introduction	4
Project Summary	5
Objectives of the Project	5
Features of the System	6
Applications of the System	6
Working of the System	7
System Requirements	9
Hardware Requirements	9
Software Requirements	9
Design and Implementation	9
Output	11
Conclusion	12
References	13

Acknowledgement

I would like to express my sincere gratitude to everyone who contributed to the successful completion of this project. I am deeply thankful to my instructor for providing guidance, support, and encouragement throughout the development of this work. Their valuable suggestions and insights helped me understand the concepts more clearly and implement them effectively.

I am also grateful to my institution for offering the necessary resources and a supportive learning environment that allowed me to explore and apply programming concepts practically. This project provided me with an opportunity to enhance my understanding of Object-Oriented Programming in C++ and its real-world applications.

Finally, I would like to thank my family and friends for their constant motivation and support during the course of this project. Their encouragement played an important role in helping me stay focused and determined.

This project has been a significant learning experience, and I am thankful to all who guided and supported me throughout the process.

Introduction:

The Number Guessing Game is a simple console-based project developed in C++. It is a fun and interactive game in which the computer randomly selects a number within a given range, and the player tries to guess that number. After each guess, the program provides feedback such as “Too high” or “Too low” to help the player reach the correct answer.

The main objective of this project is to demonstrate the use of basic programming concepts in C++, such as loops, conditional statements, functions, and random number generation. It helps beginners understand how to use the `rand()` and `srand()` functions from the C++ standard library to create random values and how to control program flow using iterative and decision-making statements.

This project not only improves logical thinking but also provides hands-on experience with user interaction and program structure in C++. It is simple yet effective in showcasing fundamental programming skills and serves as an excellent starting point for students learning game development using C++.

Project Summary

The Number Guessing Game is a simple console-based project developed in C++ where the computer randomly selects a number, and the player tries to guess it. After each guess, the program gives hints like “Too high” or “Too low” until the correct number is guessed. The project uses basic C++ concepts such as loops, conditional statements, and random number generation. It is an interactive and fun way to understand fundamental programming logic and improve problem-solving skills.

Objectives of the Project

Here are the objectives of the Number Guessing Game project:

1. To understand basic C++ concepts such as loops, conditional statements, and input/output operations.
 2. To implement random number generation using the `rand()` and `srand()` functions.
 3. To develop logical thinking and problem-solving skills through a simple and interactive game.
 4. To learn decision-making in programming by comparing user input with the computer-generated number.
 5. To create an engaging user experience through feedback messages like “Too high” or “Too low”.
-

Features of the System

The project contains several useful features:

- 1. Interactive Gameplay:** The player interacts with the computer by entering guesses and receiving instant feedback.
- 2. Random Number Generation:** The computer automatically generates a random number within a specific range (e.g., 1–100).
- 3. Hint System:** After each guess, the program provides hints like “Too high” or “Too low” to guide the player.
- 4. Multiple Rounds:** The player can choose to play the game multiple times without restarting the program.
- 5. Attempt Counter:** The game keeps track of how many attempts the player takes to guess the correct number.
- 6. User-Friendly Interface:** Simple and easy-to-understand console-based interface suitable for beginners.
- 7. Replay Option:** After finishing one round, the user can decide whether to play again or exit the game.

Applications of the System

Here are the applications of the Number Guessing Game system written in clear points:

1.Educational Use: Helps students and beginners learn and practice basic C++ programming concepts like loops, conditionals, and random number generation.

2.Logical Thinking Development:Improves problem-solving and logical reasoning skills by requiring players to think strategically while guessing.

3. Entertainment Purpose:Provides a simple and enjoyable game experience for users of all ages.

Working of the System

1. Start of the Game: When the program starts, it displays a welcome message and explains the rules of the game.

2.Random Number Generation:The computer uses the ``rand()``` function to generate a random number within a specific range (for example, between 1 and 100).

3.User Input: The player is asked to enter a guess for the number.

4.Comparison and Feedback:The program compares the player's guess with the computer-generated number:

If the guess is too high, it displays "Too high! Try again."

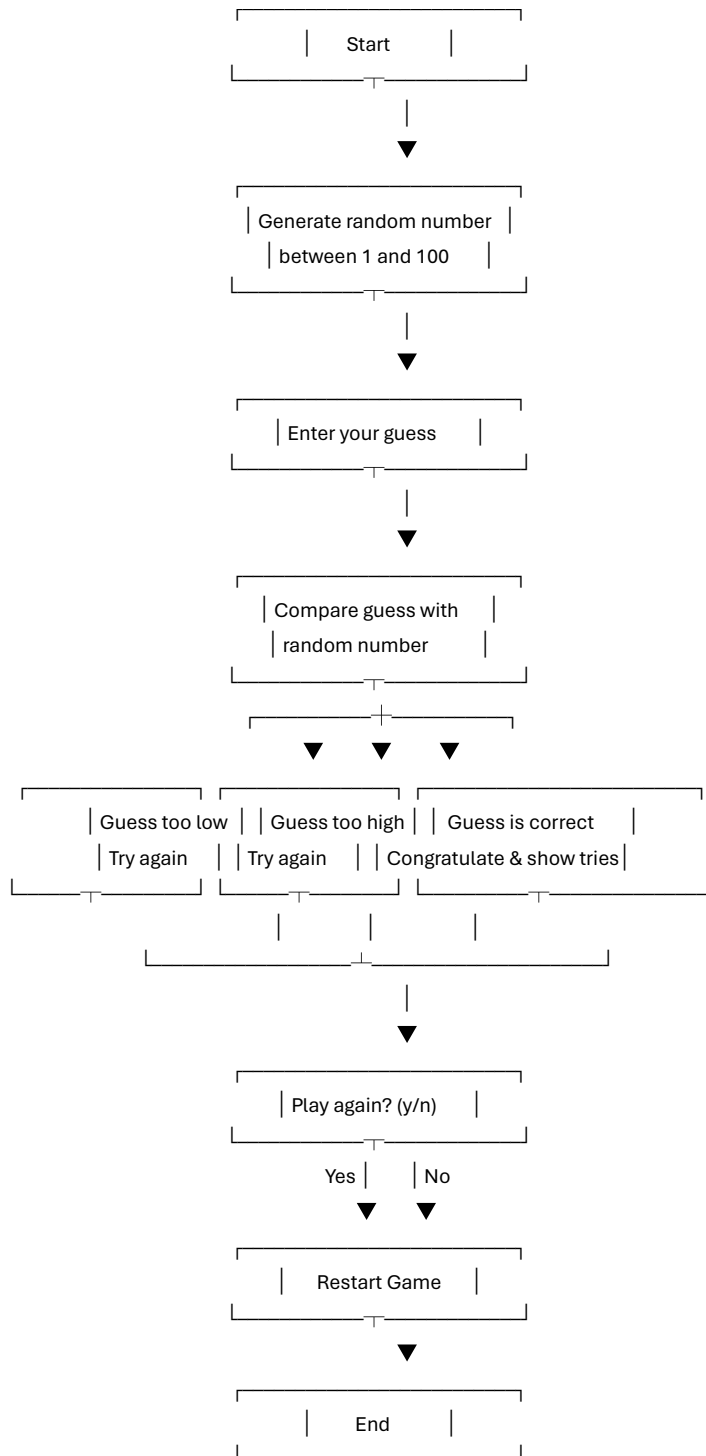
If the guess is too low, it displays "Too low! Try again."

If the guess is correct, it congratulates the player and shows the number of attempts taken.

5.Repeat Process:The game continues in a loop until the player guesses the correct number.

6.Play Again Option:After one round, the player is asked whether they want to play again. If they choose "yes," the game restarts with a new random number.

7.End of the Game:When the player chooses not to continue, the program thanks them and ends.



System Requirements

1. Hardware Requirements**

Processor: Intel Core i3 or equivalent (minimum)

RAM: 2 GB or more

Storage: 100 MB free space (for compiler and project files)

Input Devices: Keyboard and mouse

Display Any monitor with 1024x768 resolution or higher

2. Software Requirements

Operating System: Windows 7/8/10, Linux, or macOS

Compiler: C++ compiler (such as GCC, MinGW, or Code::Blocks IDE)

IDE (Optional but recommended): Code::Blocks, Dev-C++, Visual Studio, or any C++ IDE

Other Tools: Text editor (Notepad++, Sublime Text, or VS Code)

Design and Implementation

1. Design of the System

The design of the Number Guessing Game is simple and focuses on creating an interactive console application. The system design involves the following components:

User Interface (UI): A console-based interface where the user inputs guesses and receives feedback. It is text-based, simple, and user-friendly.

Game Logic:

- * Compare the player's input with the generated number.
- * Provide hints like "Too high" or "Too low."
- * Keep track of the number of attempts.
- * Repeat until the correct number is guessed.

2. Implementation:

The implementation involves translating the design into C++ code using standard programming constructs:

1. Include Required Libraries

```
```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
```
```

2. Initialize Random Number Generator

```
```cpp
srand(time(0)); // Seed for random number
int number = rand() % 100 + 1; // Number between 1-100
```

### 3. Get User Input and Compare

- \* Input is taken using `cin`.
- \* Compare with `if` and `else if` statements to give hints.

4.Count Attempts: Maintain a counter variable to track the number of guesses.

### 5. Loop Until Correct Guess:

```
```cpp
do {
    // Input guess and compare
} while (guess != number);
```

6. Replay Option:

- * After completing one round, ask the user if they want to play again (`y/n`).
- * If yes, restart; if no, terminate the program.

Output

```
🎯Welcome to the Number Guessing Game! 🎯  
  
I have chosen a number between 1 and 100.  
Can you guess it?  
Enter your guess: 50  
Too low! Try again.  
Enter your guess: 65  
🎉Congratulations! You guessed it in 2 attempts!  
  
Do you want to play again? (y/n): n  
  
Thank you for playing! Goodbye 🙌  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Conclusion:

The Number Guessing Game project demonstrates the practical application of fundamental C++ programming concepts such as loops, conditional statements, random number generation, and user input/output handling. It provides an interactive and engaging experience for the user while reinforcing logical thinking and problem-solving skills.

Through this project, beginners can understand how to design, implement, and test a simple console-based application. It also highlights the importance of program structure, control flow, and user interaction in creating functional software. Overall, this project serves as a strong foundation for learning more complex C++ programming and game development concepts.

References:

1. Deitel, P. J., & Deitel, H. M. "C++ How to Program." 10th Edition, Pearson, 2017.
2. Stroustrup, B. "The C++ Programming Language." 4th Edition, Addison-Wesley, 2013.
3. Malik, D. S. "C++ Programming: From Problem Analysis to Program Design." Cengage Learning, 2017.
4. GeeksforGeeks. "C++ Random Number Generation."
[<https://www.geeksforgeeks.org/random-number-generation-in-cpp/>]
(<https://www.geeksforgeeks.org/random-number-generation-in-cpp/>)